OPERATIONS MANAGEMENT

FORECASTING

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Forecasting

- A **forecast** is a prediction of future events used for planning purposes.

- Planning, on the other hand, is the process of making management decisions on how to deploy resources to best respond to the demand forecasts.

- Forecasting methods may be based on mathematical models that use available historical data, or on qualitative methods that draw on managerial experience and judgments, or on a combination of both.

- In this chapter, our focus is on demand forecasts. We begin with different types of demand patterns. We examine forecasting methods in three basic categories: (1) judgment, (2) causal, and (3) time-series methods.

- Forecast errors are defined, providing important clues for making better forecasts. We next consider the forecasting techniques for making forecasts and designing the forecasting system.

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- Forecasting overall demand typically originates with marketing, but internal customers throughout the organization depend on forecasts to formulate and execute their plans as well.

- Finance needs forecasts to project cash flows and capital requirements.

- Human resources uses forecasts to anticipate hiring and training needs.

- Marketing is an important source for sales forecast information because it is closest to external customers.

- Operations and supply chain managers need forecasts to plan output levels, purchases of services and materials, workforce and output schedules, inventories, and long-term capacities.

- At the supply chain level, a firm needs forecasts to coordinate with its customers and suppliers. At the process level, output forecasts are needed to design the various processes throughout the organization, including identifying and dealing with in-house bottlenecks.

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Demand Patterns

- Forecasting demand requires uncovering the underlying patterns from available information.

- The repeated observations of demand for a service or product in their order of occurrence form a pattern known as a **time series**. There are five basic patterns of most demand time series:

1. *Horizontal.* The fluctuation of data around a constant mean.

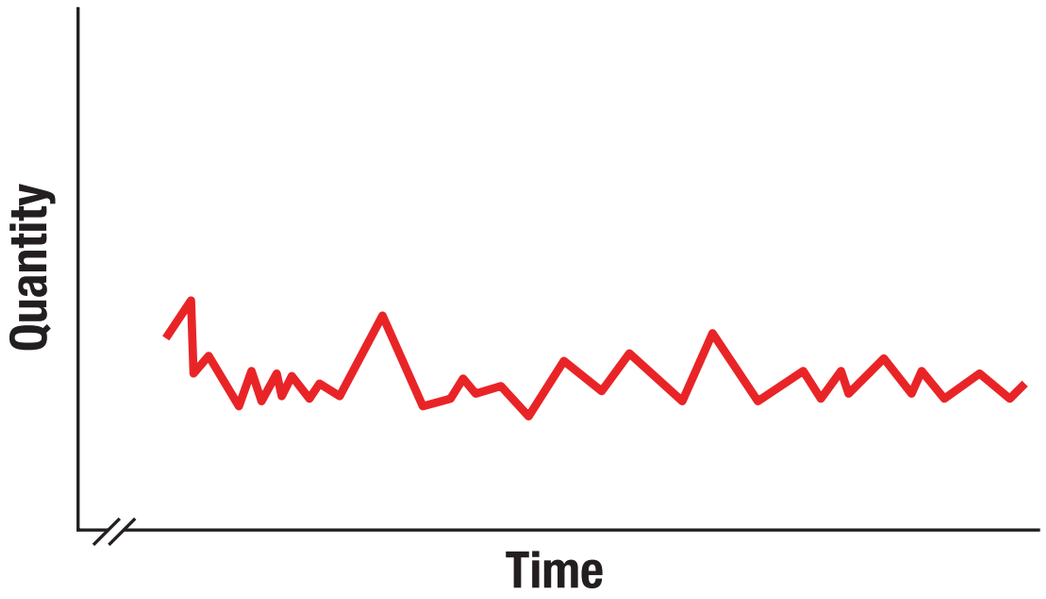
2. *Trend.* The systematic increase or decrease in the mean of the series over time.

3. *Seasonal.* A repeatable pattern of increases or decreases in demand, depending on the time of day, week, month, or season.

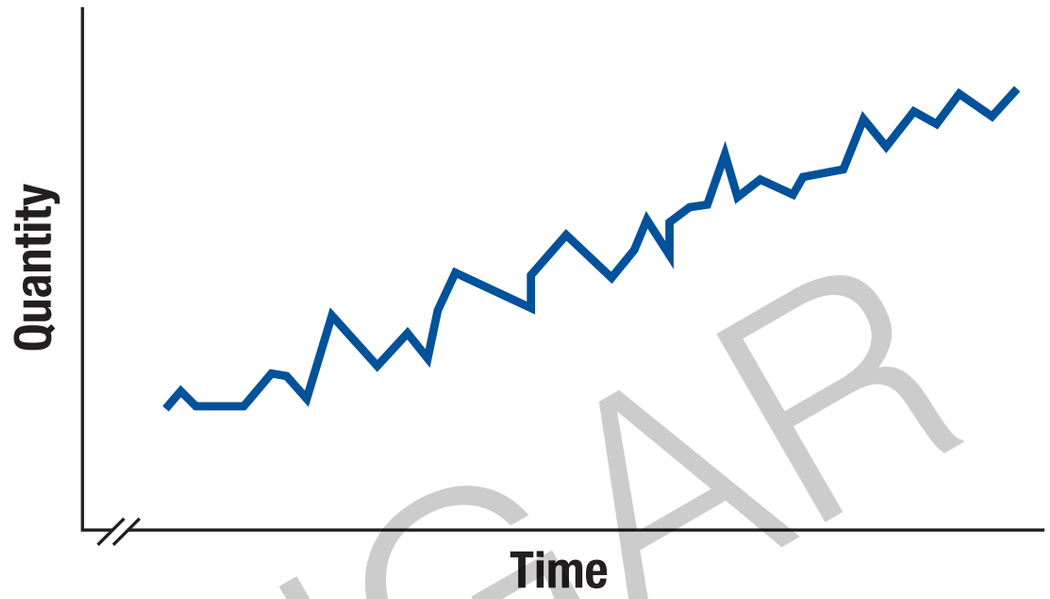
4. *Cyclical.* The less predictable gradual increases or decreases in demand over longer periods of time (years or decades).

5. *Random.* The unforecastable variation in demand.

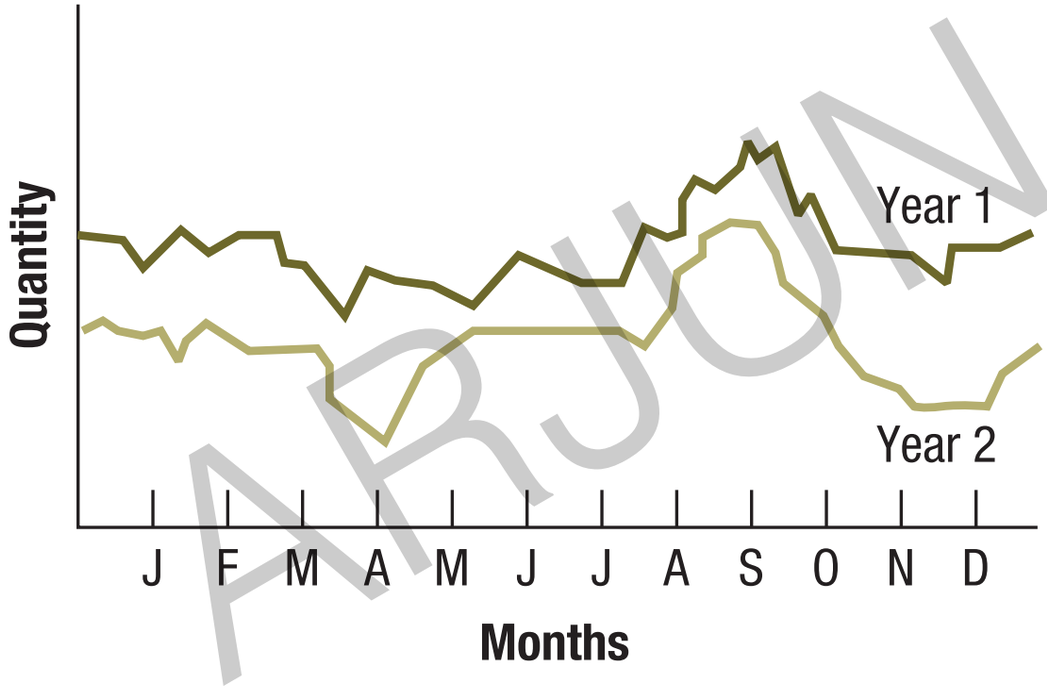
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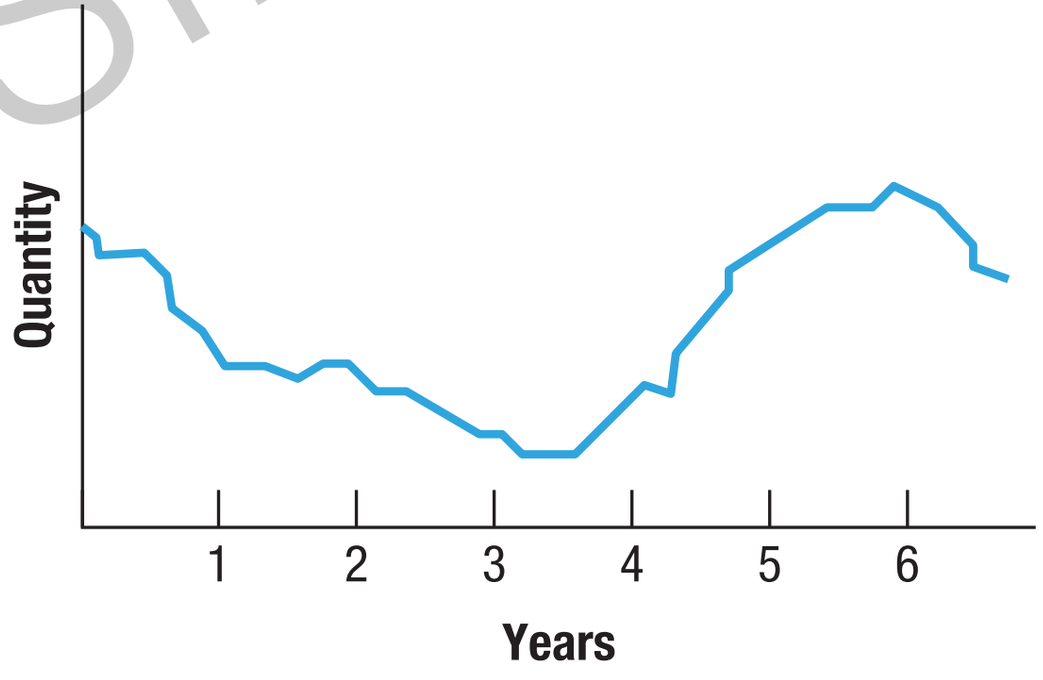
(a) Horizontal: Data cluster about a horizontal line.



(b) Trend: Data consistently increase or decrease.



(c) Seasonal: Data consistently show peaks and valleys.



(d) Cyclical: Data reveal gradual increases and decreases over extended periods of time.

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Key Decisions on Making Forecasts

- Before using forecasting techniques, a manager must make two decisions: (1) what to forecast, and (2) what type of forecasting technique to select for different items.

1. Deciding What to Forecast

- Few companies err by more than 5 percent when forecasting the annual total demand for all their services or products. However, errors in forecasts for individual items and shorter time periods may be much higher. Recognizing this reality, many companies use a two-tier forecasting system.

- **Level of Aggregation:** They first cluster (or "roll up") several similar services or products in a process called **aggregation**, making forecasts for families of services or goods that have similar demand requirements and common processing, labor, and materials requirements. Next, they derive forecasts for individual items, which are sometimes called stock-keeping units. A *stock-keeping unit (SKU)* is an individual item or product that is held in inventory somewhere along the supply chain, such as in a distribution center.

- **Units of Measurement:** Then, they select the correct unit of measurement in terms of **number of service or product units**. If accurately forecasting demand for a service or product is not possible, they forecast the **standard labor or machine-hours** required of each of the critical resources.

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2. Choosing the Type of Forecasting Technique

- Forecasting systems offer a variety of techniques, and no one of them is best for all items and situations. The forecaster's objective is to develop a useful forecast from the information at hand with the technique that is appropriate for the different patterns of demand.

- Two general types of forecasting techniques are used: **judgment methods and quantitative methods.**

1. Judgment methods translate the opinions of managers, expert opinions, consumer surveys, and salesforce estimates into quantitative estimates.

2. Quantitative methods include causal methods, time-series analysis, and trend projection with regression.

- **Causal methods** use historical data on independent variables, such as promotional campaigns, economic conditions, and competitors' actions, to predict demand.

- **Time-series analysis** is a statistical approach that relies heavily on historical demand data to project the future size of demand and recognizes trends and seasonal patterns.

- **Trend projection using regression** is a hybrid between a time-series technique and the causal method.

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Judgment Methods

- A Forecasts from quantitative methods are possible only when there is adequate historical data, (i.e., the *history file*). However, the history file may be nonexistent when a new product is introduced or when technology is expected to change. In such cases, judgment methods are the only practical way to make a forecast.

- **Salesforce estimates:** The forecasts that are compiled from estimates of future demands made periodically by members of a company's salesforce. (Regional or national sales estimates).

- **Executive opinion:** A forecasting method in which the opinions, experience, and technical knowledge of one or more managers are summarized to arrive at a single forecast. Technological forecasting is an application of executive opinion to keep abreast of the latest advances in technology.

- **Market research:** A systematic approach to determine external consumer interest in a service or product by creating and testing hypotheses through data-gathering surveys. Conducting a market research study includes designing a questionnaire, deciding how to administer it, selecting a representative sample, and analyzing the information using judgment and statistical tools to interpret the responses.

- **Delphi method:** A process of gaining consensus from a group of experts while maintaining their anonymity. This form of forecasting is useful when no historical data are available from which to develop statistical models and when managers inside the firm have no experience on which to base informed projections. A coordinator sends questions to each member of the group of outside experts, who may not even know who else is participating. The coordinator prepares a statistical summary of the responses along with a summary of arguments for particular responses. The report is sent to the same group for another round, and the participants may choose to modify their previous responses. These rounds continue until consensus is obtained.

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Example: Calculating Forecast Error Measures

The following table shows the actual sales of upholstered chairs for a furniture manufacturer and the forecasts made for each of the last 8 months. Calculate CFE, MSE, σ, MAD, and MAPE for this product.

Month, t	Demand, D_t	Forecast, F_t	Error, E_t	Error, Squared, E_t^2	Absolute Error $\lvert E_t \rvert$	Absolute Percent Error, $(\lvert E_t \rvert / D_t)(100)$
1	200	225	−25	625	25	12.5%
2	240	220	20	400	20	8.3
3	300	285	15	225	15	5.0
4	270	290	−20	400	20	7.4
5	230	250	−20	400	20	8.7
6	260	240	20	400	20	7.7
7	210	250	−40	1,600	40	19.0
8	275	240	35	1,225	35	12.7
		Total	−15	5,275	195	81.3%

SOLUTION

Using the formulas for the measures, we get

Cumulative forecast error (bias):

CFE = −15 (the bias, or the sum of the errors for all time periods in the time series)

Average forecast error (mean bias):

$$\bar{E} = \frac{\text{CFE}}{n} = \frac{-15}{8} = -1.875$$

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Mean squared error:

$$\text{MSE} = \frac{\Sigma E_t^2}{n} = \frac{5{,}275}{8} = 659.4$$

Standard deviation of the errors:

$$\sigma = \sqrt{\frac{\Sigma[E_t - (-1.875)]^2}{7}} = 27.4$$

Mean absolute deviation:

$$\text{MAD} = \frac{\Sigma|E_t|}{n} = \frac{195}{8} = 24.4$$

Mean absolute percent error:

$$\text{MAPE} = \frac{[\Sigma|E_t|/D_t]100}{n} = \frac{81.3\%}{8} = 10.2\%$$

A CFE of −15 indicates that the forecast has a slight bias to overestimate demand. The MSE, σ, and MAD statistics provide measures of forecast error variability. A MAD of 24.4 means that the average forecast error was 24.4 units in absolute value. The value of σ, 27.4, indicates that the sample distribution of forecast errors has a standard deviation of 27.4 units. A MAPE of 10.2 percent implies that, on average, the forecast error was about 10 percent of actual demand. These measures become more reliable as the number of periods of data increases.

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Causal Methods: Linear Regression

- Causal methods are used when historical data are available and the relationship between the factor to be forecasted and other external or internal factors (e.g. advertising promotions) can be identified.

- Causal methods are good for predicting turning points in demand and for preparing long-range forecasts. Here, we focus on linear regression, one of the best known and most commonly used causal methods.

- In **linear regression**, one variable, called a dependent variable, is related to one or more independent variables by a linear equation.

- The **dependent variable** (such as demand for door hinges) is the one the manager wants to forecast.

- The **independent variables** (such as advertising expenditures) are assumed to affect the dependent variable and thereby "cause" the results observed in the past.

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- In the simplest linear regression models, the dependent variable is a function of only one independent variable and, therefore, the theoretical relationship is a straight line:

$$Y = a + bX$$

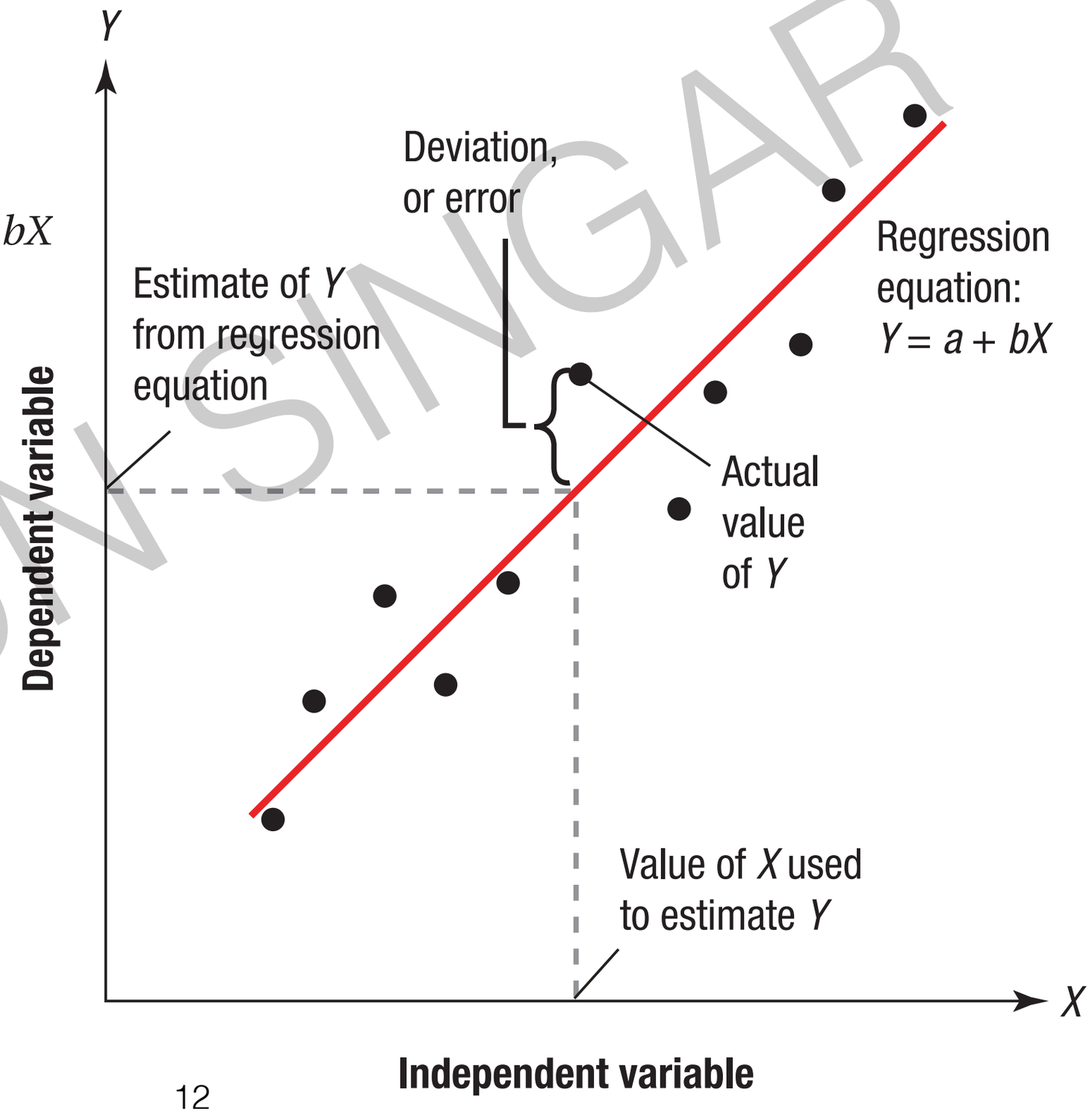
where

$Y =$ dependent variable

$X =$ independent variable

$a =$ Y-intercept of the line

$b =$ slope of the line



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- The objective of linear regression analysis is to find values of a and b and provide measures of forecast accuracy.

- Three measures commonly reported are (1) the sample correlation coefficient, (2) the sample coefficient of determination, and (3) the standard error of the estimate.

- The *sample correlation coefficient, r*, measures the direction and strength of the relationship between the independent variable and the dependent variable. The value of *r* can range from -1.00 to +1.00.

- The *sample coefficient of determination* measures the amount of variation in the dependent variable about its mean that is explained by the regression line. The coefficient of determination is the square of the correlation coefficient, or *r sq*. The value of *r sq rang*es from 0.00 to 1.00. Regression equations with a value of *r sq* close to 1.00 mean a close fit.

- The *standard error of the estimate*, s_{xy}, measures how closely the data on the dependent variable cluster around the regression line. Although it is similar to the sample standard deviation, it measures the error from the dependent variable, *Y*, to the regression line, rather than to the mean. Thus, it is the standard deviation of the difference between the actual demand and the estimate provided by the regression equation.

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Example: Using Linear Regression to Forecast Product Demand

The supply chain manager seeks a better way to forecast the demand for door hinges and believes that the demand is related to advertising expenditures. The following are sales and advertising data for the past 5 months:

Month	Sales (Thousands of Units)	Advertising (Thousands of $)
1	264	2.5
2	116	1.3
3	165	1.4
4	101	1.0
5	209	2.0

The company will spend $1,750 next month on advertising for the product. Use linear regression to develop an equation and a forecast for this product.

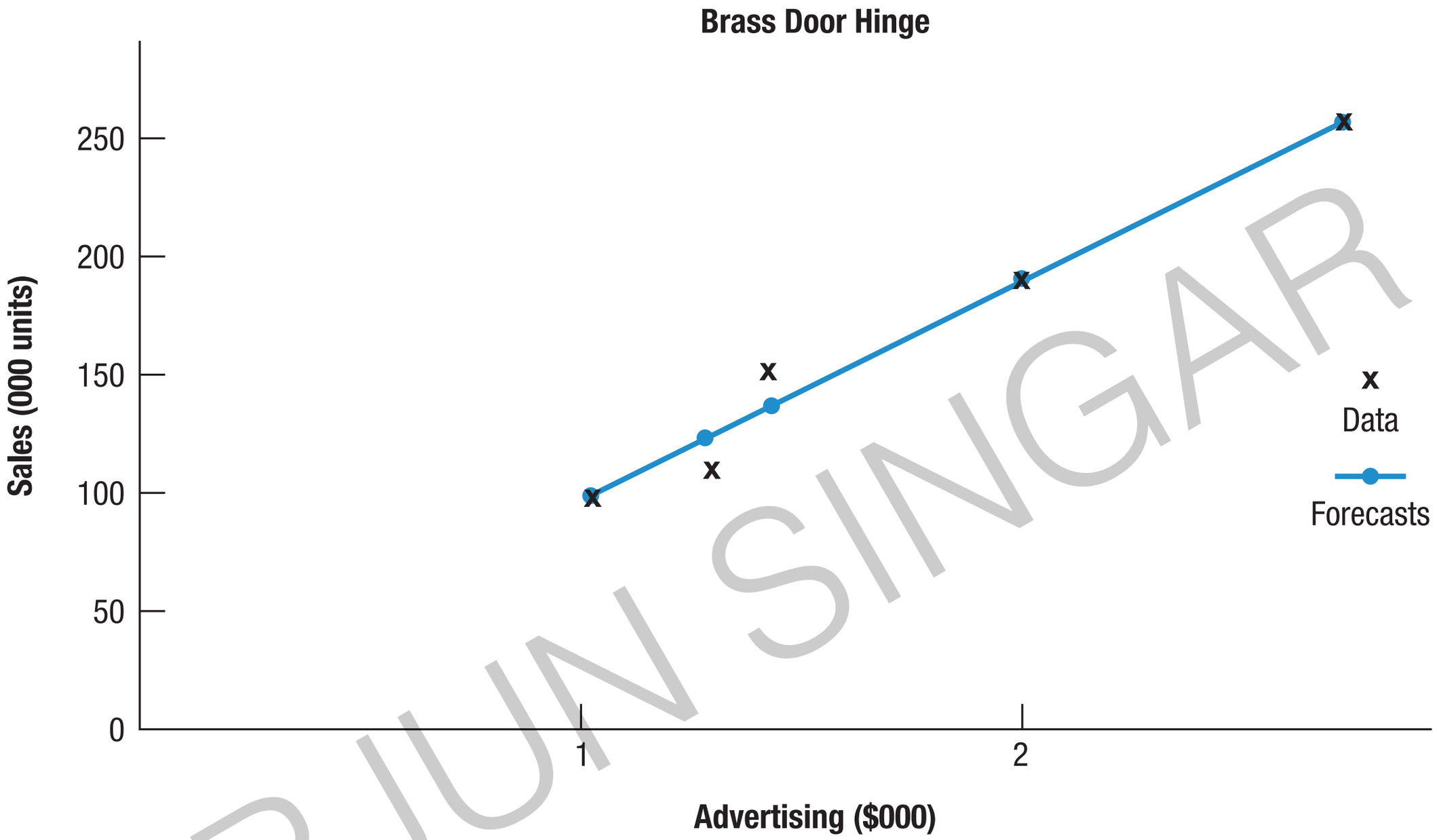
$$a = -8.135$$
$$b = 109.229X$$
$$r = 0.980$$
$$r^2 = 0.960$$
$$s_{yx} = 15.603$$

The regression equation is

$$Y = -8.135 + 109.229X$$

and the regression line is shown in Figure 14.4. The sample correlation coefficient, r, is 0.98, which is unusually close to 1.00 and suggests an unusually strong positive relationship exists between sales and advertising expenditures. The sample coefficient of determination, r^2, implies that 96 percent of the variation in sales is explained by advertising expenditures.

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DECISION POINT

The supply chain manager decided to use the regression model as input to planning production levels for month 6. As the advertising expenditure will be $1,750, the forecast for month 6 is $Y = -8.135 + 109.229(1.75) = 183.016$, or 183,016 units.

Time-Series Methods

- Rather than using independent variables for the forecast as regression models do, time-series methods use historical information regarding only the dependent variable.

- These methods are based on the assumption that the dependent variable's past pattern will continue in the future.

- Time-series analysis identifies the underlying patterns of demand that combine to produce an observed historical pattern of the dependent variable and then develops a model to replicate it.

- In this section, we focus on time-series methods that address the horizontal, trend, and seasonal patterns of demand.

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Naïve Forecast

- A method often used in practice is the **naïve forecast**, whereby the forecast for the next period (F_{t+1}) equals the demand for the current period (D_t). So if the actual demand for Wednesday is 35 customers, the forecasted demand for Thursday is 35 customers.

- The naïve forecast method may be adapted to take into account a demand trend. The increase (or decrease) in demand observed between the last two periods is used to adjust the current demand to arrive at a forecast.

- Suppose that last week the demand was 120 units and the week before it was 108 units. Demand increased 12 units in 1 week, so the forecast for next week would be 120 + 12 = 132 units.

- The naïve forecast method also may be used to account for seasonal patterns. If the demand last July was 50,000 units, and assuming no underlying trend from one year to the next, the forecast for this July would be 50,000 units.

- The method works best when the horizontal, trend, or seasonal patterns are stable and random variation is small.

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Estimating the Average

- We begin our discussion of statistical methods of time-series forecasting with demand that has no apparent trend, seasonal, or cyclical patterns. The horizontal pattern in a time series is based on the mean of the demands, so we focus on forecasting methods that estimate the average of a time series of data.

- The forecast of demand for *any* period in the future is the average of the time series computed in the current period. For example, if the average of past demand calculated on Tuesday is 65 customers, the forecasts for Wednesday, Thursday, and Friday are 65 customers each day. Assuming that the time series has only a horizontal and random pattern, one approach is simply to calculate the average of the data.

- However, this approach has no adaptive quality if there is a trend, seasonal, or cyclical pattern. The statistical techniques that do have an adaptive quality in estimating the average in a time series are (1) simple moving averages, (2) weighted moving averages, and (3) exponential smoothing.

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- **Simple Moving Average Method:** A time-series method used to estimate the average of a demand time series by averaging the demand for the *n* most recent time periods.

$$F_{t+1} = \frac{\text{Sum of last } n \text{ demands}}{n} = \frac{D_t + D_{t-1} + D_{t-2} + \cdots + D_{t-n+1}}{n}$$

- **Weighted Moving Average Method:** A time-series method in which each historical demand in the average can have its own weight; the sum of the weights equals 1.0.

$$F_{t+1} = 0.50D_t + 0.30D_{t-1} + 0.20D_{t-2}$$

- **Exponential Smoothing Method:** A weighted moving average method that calculates the average of a time series by implicitly giving recent demands more weight than earlier demands.

$$F_{t+1} = \alpha D_t + (1 - \alpha)F_t$$

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Example: Using the Moving Average Method to Estimate Average Demand

a. Compute a *three-week* moving average forecast for the arrival of medical clinic patients in week 4. The numbers of arrivals for the past 3 weeks were as follows:

Week	Patient Arrivals
1	400
2	380
3	411

b. If the actual number of patient arrivals in week 4 is 415, what is the forecast error for week 4?

c. What is the forecast for week 5?

SOLUTION

a. The moving average forecast at the end of week 3 is

$$F_4 = \frac{411 + 380 + 400}{3} = 397.0$$

b. The forecast error for week 4 is

$$E_4 = D_4 - F_4 = 415 - 397 = 18$$

c. The forecast for week 5 requires the actual arrivals from weeks 2 through 4, the 3 most recent weeks of data.

$$F_5 = \frac{415 + 411 + 380}{3} = 402.0$$

DECISION POINT

Thus, the forecast at the end of week 3 would have been 397 patients for week 4, which fell short of actual demand by 18 patients. The forecast for week 5, made at the end of week 4, would be 402 patients. If a forecast is needed now for week 6 and beyond, it would also be for 402 patients.

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Example: Using the Weighted Moving Average Method to Estimate Average Demand

The Polish General's Pizza Parlor is a small restaurant catering to patrons with a taste for European pizza. One of its specialties is Polish Prize pizza. The manager must forecast weekly demand for these special pizzas so that he can order pizza shells weekly. Recently, demand has been as follows:

Week	Pizzas	Week	Pizzas
June 2	50	June 23	56
June 9	65	June 30	55
June 16	52	July 7	60

a. Forecast the demand for pizza for June 23 to July 14 by using the simple moving average method with $n = 3$. Then, repeat the forecast by using the weighted moving average method with $n = 3$ and weights of 0.50, 0.30, and 0.20, with 0.50 applying to the most recent demand.

b. Calculate the MAD for each method.

SOLUTION

a. The simple moving average method and the weighted moving average method give the following results:

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Current Week	Simple Moving Average Forecast for Next Week	Weighted Moving Average Forecast for Next Week
June 16	$\frac{52 + 65 + 50}{3} = 55.7$ or 56	$[(0.5 \times 52) + (0.3 \times 65) + (0.2 \times 50)] = 55.5$ or 56
June 23	$\frac{56 + 52 + 65}{3} = 55.7$ or 58	$[(0.5 \times 56) + (0.3 \times 52) + (0.2 \times 65)] = 56.6$ or 57
June 30	$\frac{55 + 56 + 52}{3} = 54.3$ or 54	$[(0.5 \times 55) + (0.3 \times 56) + (0.2 \times 52)] = 54.7$ or 55
July 7	$\frac{60 + 55 + 56}{3} = 57.0$ or 57	$[(0.5 \times 60) + (0.3 \times 55) + (0.2 \times 56)] = 57.7$ or 58

b. The mean absolute deviation is calculated as follows:

		SIMPLE MOVING AVERAGE		WEIGHTED MOVING AVERAGE	
Week	Actual Demand	Forecast for This Week	Absolute Errors $\lvert E_t \rvert$	Forecast for This Week	Absolute Errors $\lvert E_t \rvert$
June 23	56	56	$\lvert 56 - 56 \rvert = 0$	56	$\lvert 56 - 56 \rvert = 0$
June 30	55	58	$\lvert 55 - 58 \rvert = 3$	57	$\lvert 55 - 57 \rvert = 2$
July 7	60	54	$\lvert 60 - 54 \rvert = 6$	55	$\lvert 60 - 55 \rvert = 5$
			$\text{MAD} = \frac{0 + 3 + 6}{3} = 3.0$		$\text{MAD} = \frac{0 + 2 + 5}{3} = 2.3$

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Example: Using Exponential Smoothing Method to Estimate Average Demand

a. Reconsider the patient arrival data in Example 14.3. It is now the end of week 3, so the actual number of arrivals is known to be 411 patients. Using $\alpha = 0.10$, calculate the exponential smoothing forecast for week 4.

b. What was the forecast error for week 4 if the actual demand turned out to be 415?

c. What is the forecast for week 5?

SOLUTION

a. The exponential smoothing method requires an initial forecast. Suppose that we take the demand data for the first 2 weeks and average them, obtaining $(400 + 380)/2 = 390$ as an initial forecast. (POM for Windows and OM Explorer simply use the actual demand for the first week as a default setting for the initial forecast for period 1, and do not begin tracking forecast errors until the second period). To obtain the forecast for week 4, using exponential smoothing with $D_3 = 411$, $\alpha = 0.10$, and $F_3 = 390$, we calculate the forecast for week 4 as

$$F_4 = 0.10(411) + 0.90(390) = 392.1$$

Thus, the forecast for week 4 would be 392 patients.

b. The forecast error for week 4 is

$$E_4 = 415 - 392 = 23$$

c. The new forecast for week 5 would be

$$F_5 = 0.10(415) + 0.90(392.1) = 394.4$$

or 394 patients. Note that we used F_4, not the integer-value forecast for week 4, in the computation for F_5. In general, we round off (when it is appropriate) only the final result to maintain as much accuracy as possible in the calculations.

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Trend Projection with Regression

- Trend projection with regression is a forecasting model that accounts for the trend with simple regression analysis. To develop a regression model for forecasting the trend, let the dependent variable, *Y*, be a period's demand and the independent variable, *t*, be the time period. For the first period, let *t* = 1; for the second period, let *t* = 2; and so on. The regression equation is $F_t = a + bt$.

- One advantage of the trend projection with regression model is that it can forecast demand well into the future. The previous models project demand just one period ahead, and assume that demand beyond that will remain at that same level.

- One apparent disadvantage of the trend with regression model is that it is not adaptive. The solution to this problem comes when you answer the following question. If you had the past sales of Ford automobiles since 1920, would you include each year in your regression analysis, giving equal weight to each year's sales, or include just the sales for more recent years? You most likely would decide to include just the more recent years, making your regression model more adaptive. The trend projection with regression model can thus be made more or less adaptive by the selection of historical data periods to include in the same way that moving average (changing *n*) or exponential smoothing (changing alpha) models do.

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Example: Using Trend Projection with Regression to Forecast a Demand Series with a Trend

Medanalysis, Inc., provides medical laboratory services to patients of Health Providers, a group of 10 family-practice doctors associated with a new health maintenance program. Managers are interested in forecasting the number of blood analysis requests per week. Recent publicity about the damaging effects of cholesterol on the heart has caused a national increase in requests for standard blood tests. The arrivals over the last 16 weeks are given in Table 14.1. What is the forecasted demand for the next three periods?

TABLE 14.1 ARRIVALS AT MEDANALYSIS FOR LAST 16 WEEKS

Week	Arrivals	Week	Arrivals
1	28	9	61
2	27	10	39
3	44	11	55
4	37	12	54
5	35	13	52
6	53	14	60
7	38	15	60
8	57	16	75

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				CFE	Averages MSE	MAD	MAPE
				0.000	**52.958**	**6.210**	**13.53%**
Period #	Actual Demand	Forecast	Error	Running CFE	Error Squared	Absolute Error	Abs % error
1	28	31	-2.846	-2.846	8.097	2.846	10.16%
2	27	33	-6.191	-9.037	38.331	6.191	22.93%
3	44	36	8.463	-0.574	71.626	8.463	19.23%
4	37	38	-0.882	-1.456	0.779	0.882	2.38%
5	35	40	-5.228	-6.684	27.331	5.228	14.94%
6	53	43	10.426	3.743	108.711	10.426	19.67%
7	38	45	-6.919	-3.176	47.874	6.919	18.21%
8	57	47	9.735	6.559	94.776	9.735	17.08%
9	61	50	11.390	17.949	129.725	11.390	18.67%
10	39	52	-12.956	4.993	167.855	12.956	33.22%
11	55	54	0.699	5.691	0.488	0.699	1.27%
12	54	57	-2.647	3.044	7.007	2.647	4.90%
13	52	59	-6.993	-3.949	48.897	6.993	13.45%
14	60	61	-1.338	-5.287	1.791	1.338	2.23%
15	60	64	-3.684	-8.971	13.571	3.684	6.14%
16	75	66	8.971	0.000	80.471	8.971	11.96%

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a (Y intercept)	28.50
b (slope or trend)	2.35
r2	0.69
CFE	0.00
MAD	6.21
MSE	52.96
MAPE	13.53%
Forecast for period 17	68.375
Forecast for period 18	70.72059
Forecast for period 19	73.06618

Looking at the Results sheet of the below Figure, we see that the *Y* intercept of the trend line (*a*) is 28.50 and the slope of the line (*b*) is 2.35. Thus, the trend equation is $F_t = a + bt$, where *t* is the time period for which you are forecasting. The forecast for period 19 is 28.5 + 2.35 (19) = 73. The error terms are CFE = 0 (which is to be expected when the regression begins at the same time that error analysis begins), MAD = 6.21, MSE = 52.96, and MAPE = 13.53 percent. The coefficient of determination r^2 is decent at 0.69. The trend line is rising gently and reaches 73 for period 19. Each period the forecast predicts an increase of 2.35 arrivals per week.

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Seasonal Patterns

- Trend Seasonal patterns are regularly repeating upward or downward movements in demand measured in periods of less than one year (hours, days, weeks, months, or quarters). In this context, the time periods are called *seasons*. For example, customer arrivals at a fast-food shop on any day may peak between 11 am and 1 pm and again from 5 pm to 7 pm.

- An easy way to account for seasonal effects is to use one of the techniques already described, but to limit the data in the time series to those time periods in the same season. For example, for a day-of-the-week seasonal effect, one time series would be for Mondays, one for Tuesdays, and so on. Such an approach accounts for seasonal effects, but has the disadvantage of discarding considerable information on past demand.

- Other methods are available that analyze all past data, using one model to forecast demand for all of the seasons. We describe only the **multiplicative seasonal method**, whereby an estimate of average demand is multiplied by seasonal factors to arrive at a seasonal forecast.

- An alternative to the multiplicative seasonal method is the **additive seasonal method**, whereby seasonal forecasts are generated by adding or subtracting a seasonal constant (say, 50 units) to the estimate of average demand per season. This approach is based on the assumption that the seasonal pattern is constant, regardless of average demand. The amplitude of the seasonal adjustment remains the same regardless of the level of demand.

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Example: Using the Multiplicative Seasonal Method to Forecast the Number of Customers

The manager of the Stanley Steemer carpet cleaning company needs a quarterly forecast of the number of customers expected next year. The carpet cleaning business is seasonal, with a peak in the third quarter and a trough in the first quarter. The manager wants to forecast customer demand for each quarter of year 5, based on an estimate of total year 5 demand of 2,600 customers.

SOLUTION

The following table calculates the seasonal factor for each week.

It shows the quarterly demand data from the past 4 years, as well as the calculations performed to get the average seasonal factor for each quarter.

	YEAR 1		YEAR 2	
Quarter	Demand	Seasonal Factor (1)	Demand	Seasonal Factor (2)
1	45	45/250 = 0.18	70	70/300 = 0.23333
2	335	335/250 = 1.34	370	370/300 = 1.23333
3	520	520/250 = 2.08	590	590/300 = 1.96667
4	100	100/250 = 0.40	170	170/300 = 0.56667
Total	1,000		1,200	
Average	1,000/4 = 250		1,200/4 = 300	

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YEAR 3		YEAR 4		
Demand	**Seasonal Factor (3)**	**Demand**	**Seasonal Factor (4)**	**Average Seasonal Factor [(1+2+3+4+)/4]**
100	100/450 = 0.22222	100	100/550 = 0.18182	0.2043
585	585/450 = 1.30	725	725/550 = 1.31818	1.2979
830	830/450 = 1.84444	1160	1160/550 = 2.10909	2.0001
285	285/450 = 0.63333	215	215/550 = 0.39091	0.4977
1,800		2,200		
1,800 = 450		2,200/4 = 550		

- For example, the seasonal factor for quarter 1 in year 1 is calculated by dividing the actual demand (45) by the average demand for the whole year (1000/4 = 250). When this is done for all 4 years, we then can average the seasonal factors for quarter 1 over all 4 years. The result is a seasonal factor of 0.2043 for quarter 1.

- Once seasonal factors are calculated for all four seasons (see last column in the table on the previous page), we then turn to making the forecasts for year 5. The manager suggests a forecast of 2,600 customers for the whole year, which seems reasonable given that the annual demand has been increasing by an average of 400 customers each year (from 1,000 in year 1 to 2,200 in year 4, or 1,200/3 = 400). The computed forecast demand is found by extending that trend, and projecting an annual demand in year 5 of 2,200 + 400 = 2,600 customers.

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- The quarterly forecasts are straight-forward. First, find the average demand forecast for year 5, which is 2,600/4 = 650. Then multiply this average demand by the average seasonal index, giving us

Quarter	Forecast
1	$650 \times 0.2043 = 132.795$
2	$650 \times 1.2979 = 843.635$
3	$650 \times 2.0001 = 1{,}300.065$
4	$650 \times 0.4977 = 323.505$

- Using this seasonal method, the analyst makes a demand forecast as low as 133 customers in the first quarter and as high as 1,300 customers in the third quarter. The season of the year clearly makes a difference.

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Criteria for Selecting Time-Series Methods

- Forecast error measures provide important information for choosing the best forecasting method for a service or product. They also guide managers in selecting the best values for the parameters needed for the method: *n* for the moving average method, the weights for the weighted moving average method, alpha for the exponential smoothing method, and when regression data begins for the trend projection with regression method.

- The criteria to use in making forecast method and parameter choices include (1) minimizing bias (CFE); (2) minimizing MAPE, MAD, or MSE; (3) maximizing *r sq*; (4) meeting managerial expectations of changes in the components of demand; and (5) minimizing the forecast errors in recent periods.

- The first three criteria relate to statistical measures based on historical performance, the fourth reflects expectations of the future that may not be rooted in the past, and the fifth is a way to use whatever method seems to be working best at the time a forecast must be made.

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- **Using Statistical Criteria:** Statistical performance measures can be used in the selection of which forecasting method to use. The following guidelines will help when searching for the best time- series models:

1. For projections of more stable demand patterns, use lower alpha values or larger *n* values to emphasize historical experience.

2. For projections of more dynamic demand patterns, try higher alpha values or smaller *n* values. When historical demand patterns are changing, recent history should be emphasized.

Holdout Sample: Actual demands from the more recent time periods in the time series that are set aside to test different models developed from the earlier time periods. Once the model is selected, it is tested again but now for the holdout sample.

- **Tracking Signal:** A **tracking signal** is a measure that indicates whether a method of forecasting is accurately predicting actual changes in demand. The tracking signal measures the number of MADs represented by the cumulative sum of forecast errors, the CFE. The CFE tends to be close to 0 when a correct forecasting system is being used. At any time, however, random errors can cause the CFE to be a nonzero number. The tracking signal formula is:

$$\text{Tracking signal} = \frac{\text{CFE}}{\text{MAD}}$$

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Using Multiple Techniques for Forecasting

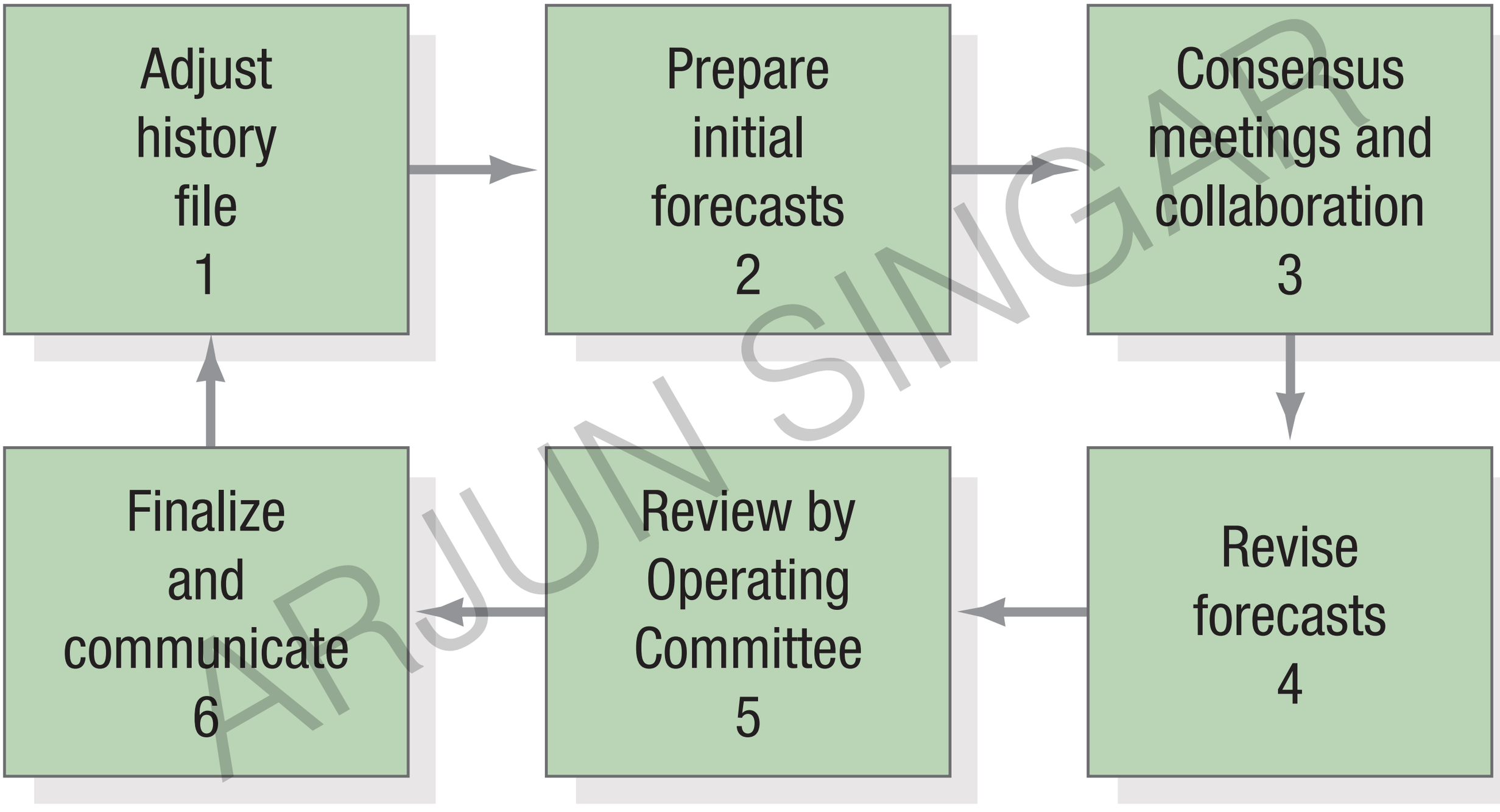
- **Combination forecasts** are forecasts that are produced by averaging independent forecasts based on different methods, different sources, or different data. It is intriguing that combination forecasts often perform better over time than even the *best* single forecasting procedure. For example, suppose that the forecast for the next period is 100 units from technique 1 and 120 units from technique 2 and that technique 1 has provided more accurate forecasts to date. The combination forecast for next period, giving equal weight to each technique, is 110 units (or 0.5 * 100 + 0.5 * 120).

- **Judgmental adjustment** is an adjustment made to forecasts from one or more quantitative models that accounts for recognizing which models are performing particularly well in recent past, or take into account contextual information.

- **Focus forecasting** is a method of forecasting that selects the best forecast from a group of forecasts generated by individual techniques. Using the history file as the starting point for each method, the computer generates forecasts for the current period. The forecasts are compared to actual demand, and the method that produces the forecast with the least error is used to make the forecast for the next period. The method used for each item may change from period to period.

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A Typical Forecasting Process



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TABLE 14.2 | SOME PRINCIPLES FOR THE FORECASTING PROCESS

- Better processes yield better forecasts.
- Demand forecasting is being done in virtually every company, either formally or informally. The challenge is to do it well—better than the competition.
- Better forecasts result in better customer service and lower costs, as well as better relationships with suppliers and customers.
- The forecast can and must make sense based on the big picture, economic outlook, market share, and so on.
- The best way to improve forecast accuracy is to focus on reducing forecast error.
- Bias is the worst kind of forecast error; strive for zero bias.
- Whenever possible, forecast at more aggregate levels. Forecast in detail only where necessary.
- Far more can be gained by people collaborating and communicating well than by using the most advanced forecasting technique or model.

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